



ASSESSMENT COVERSHEET

Trimester 2620

Attach this coversheet as the cover of your submission. All sections must be completed.

Section A: Submission Details

Programme : Bachelor of Computer Science
Course Code & Name : LDCW6123 - FUNDAMENTALS OF DIGITAL COMPETENCE FOR PROGRAMMER
Course Lecturer(s) : Mr. V Segaran A/L Veeraya
Submission Title : Group 13 Netflix Innovation Life Cycle Poster and Interactive Program
Deadline : Day 02 Month 10 Year 2026 Time 23:59
Penalties :

- Plagiarised work is an Academic Offence in University Rules & Regulations and will be penalised accordingly.
- Failure to fulfil the limits set for plagiarism similarity index and AI-generated content will result in the **essay being awarded a score not exceeding 50%** of the total score.
- Without Turnitin reports submission is deemed **incomplete**.

Section B: Academic Integrity

Tick (✓) each box below if you agree:

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|--------------------------|--|
| ☐ | I have read and understood the policy on Plagiarism in University Rules & Regulations. |
| ☐ | This submission is my own, unless indicated with proper referencing. |
| ☐ | This submission has not been previously submitted or published. |
| ☐ | This submission follows the requirements stated in the course. |

Section C: Submission Details (Group 13)

Student ID(s)	Student Name(s)	Designation
261UC240LY	Aziel Tan Zheng Chuan	Group leader
261UC240PJ	See Wing Kit	Member
261UC26145	Soo Kian Rong	Member
261UC2406W	Vincent Lock Chun Kit	Member
253UC255P1	Wong Kee Yuan	Member
262UC250T8	Ho Ming Hao	Member

Declaration by group leader

I hereby declare that all group members' names are correctly included in the above section. I hold a copy of this assignment which I can produce if the original is lost or damaged. I certify that no part of this assignment has been copied from any other student's work or from any other source except where due acknowledgement is made in the assignment.

Group leader's signature : _____

Group leader's name: Aziel Tan Zheng Chuan

Group leader's student ID: 261UC240LY

Date : _____

Group member declaration

(Each group member, including the **group leader**, must individually fill up and submit this form)

For the purpose of completing this assignment, I have performed the following tasks (please list):

I hereby declare that I have assessed this submission and I take full responsibility should there be any inaccuracies, incompleteness, omissions, delays or non-submission.

Group member's signature : _____

Group member's name: Aziel Tan Zheng Chuan

Group member's ID: 261UC240LY

Date : _____

Group member declaration

(Each group member, including the **group leader**, must individually fill up and submit this form)

For the purpose of completing this assignment, I have performed the following tasks (please list):

I hereby declare that I have assessed this submission and I take full responsibility should there be any inaccuracies, incompleteness, omissions, delays or non-submission.

Group member's signature : _____

Group member's name: See Wing Kit

Group member's ID: 261UC240PJ

Date : _____

Group member declaration

(Each group member, including the **group leader**, must individually fill up and submit this form)

For the purpose of completing this assignment, I have performed the following tasks (please list):

I hereby declare that I have assessed this submission and I take full responsibility should there be any inaccuracies, incompleteness, omissions, delays or non-submission.

Group member's signature : _____

Group member's name: Soo Kian Rong

Group member's ID: 261UC26145

Date : _____

Group member declaration

(Each group member, including the **group leader**, must individually fill up and submit this form)

For the purpose of completing this assignment, I have performed the following tasks (please list):

I hereby declare that I have assessed this submission and I take full responsibility should there be any inaccuracies, incompleteness, omissions, delays or non-submission.

Group member's signature : _____

Group member's name: Vincent Lock Chun Kit

Group member's ID: 261UC2406W

Date : _____

Group member declaration

(Each group member, including the **group leader**, must individually fill up and submit this form)

For the purpose of completing this assignment, I have performed the following tasks (please list):

I hereby declare that I have assessed this submission and I take full responsibility should there be any inaccuracies, incompleteness, omissions, delays or non-submission.

Group member's signature : _____

Group member's name: Wong Kee Yuan

Group member's ID: 253UC255P1

Date : _____

Group member declaration

(Each group member, including the **group leader**, must individually fill up and submit this form)

For the purpose of completing this assignment, I have performed the following tasks (please list):

I hereby declare that I have assessed this submission and I take full responsibility should there be any inaccuracies, incompleteness, omissions, delays or non-submission.

Group member's signature : _____

Group member's name: Ho Ming Hao

Group member's ID: 262UC250T8

Date : _____

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Netflix Innovation Life Cycle Poster and Movie Discovery Assistant

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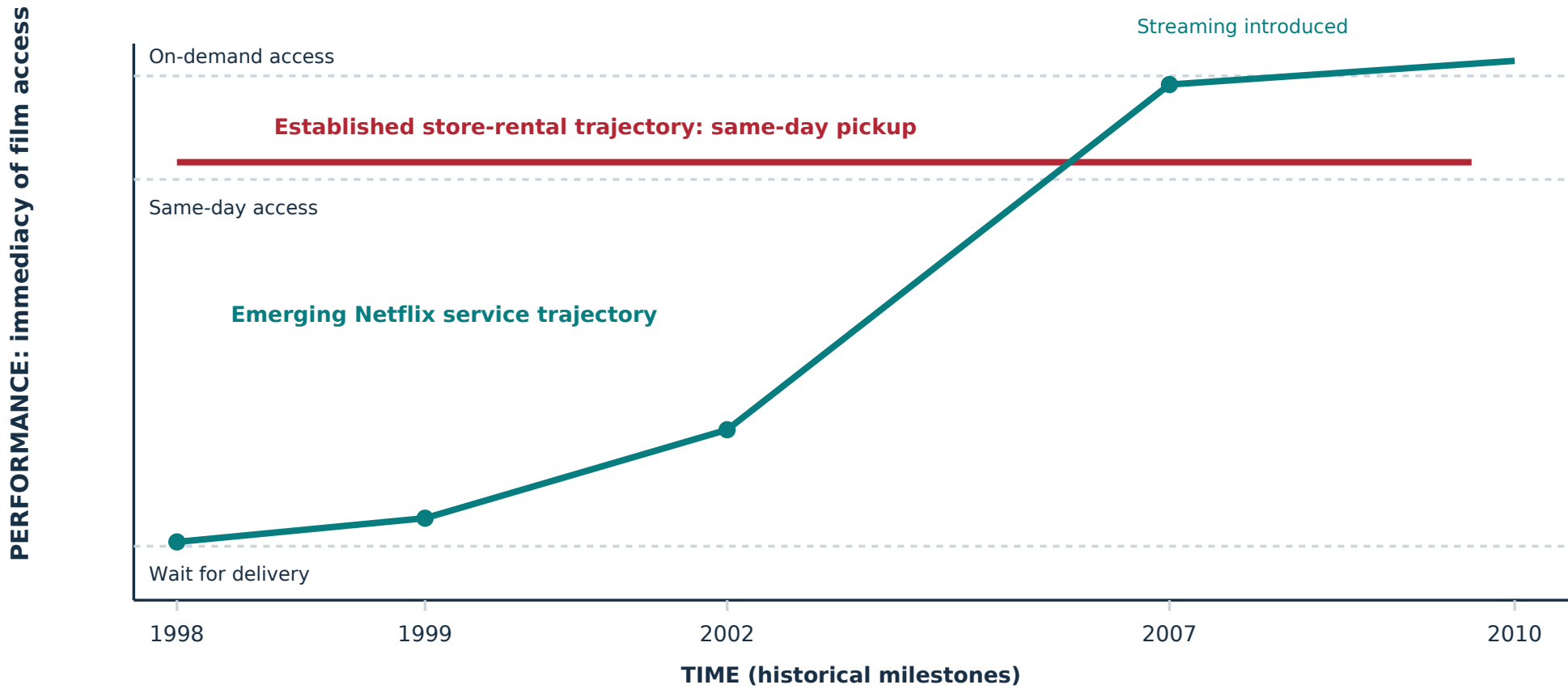
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Part 1 examines the DVD-by-mail innovation and its later development. Part 2 demonstrates movie discovery with a transparent C++ preference rule. The references and evidence support the analysis and programme behaviour.

NETFLIX AND VIDEO RENTAL DISRUPTION

Christensen model / Group 13 / Market: physical home video rental

Entrant: online DVD-by-mail. Incumbent: Blockbuster stores. The original foothold is separated from later streaming.



Conceptual schematic only. Heights are qualitative, not measured scores. No market-share values or causal estimates are plotted.

1998	1999	2002	2007	2010 / 2023
Online DVD rental Netflix, Inc., 2002a	Subscription launch Netflix, Inc., 2003	Faster regional delivery Netflix, Inc., 2002b	Streaming introduced Netflix, Inc., 2019	Bankruptcy / DVD closure Institute, n.d.; Sarandos, 2023

REFERENCES

Christensen Institute. (n.d.). Disruptive innovation theory.
<https://www.christenseninstitute.org/theory/disruptive-innovation/>

Netflix, Inc. (2002a, May 22). Netflix announces initial public offering.
<https://ir.netflix.net/investor-news-and-events/financial-releases/press-release-details/2002/Netflix-Announces-Initial-Public-Offering/default.aspx>

Netflix, Inc. (2002b, June 20). Netflix announces opening of 10 regional distribution centers.
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Netflix, Inc. (2003). Annual report on Form 10-K for the fiscal year ended December 31, 2002. U.S. Securities and Exchange Commission.
<https://www.sec.gov/Archives/edgar/data/1065280/000095016803001155/d10k.htm>

Netflix, Inc. (2019). Annual report on Form 10-K for the fiscal year ended December 31, 2018.
https://s22.q4cdn.com/959853165/files/doc_financials/annual_reports/2018/Form-10K_Q418_Filed.pdf

Sarandos, T. (2023, April 18). Netflix DVD - The final season. Netflix.
<https://about.netflix.com/en/news/netflix-dvd-the-final-season>

Part 1 Model application

Scope and model

This report examines Netflix's original online DVD-by-mail service relative to physical video-rental stores, especially Blockbuster. The question is how a service that initially required waiting could attract customers and become a wider competitive alternative. Christensen's disruptive innovation model is appropriate because the explanation follows a changing competitive process, rather than treating a new technology as automatically disruptive (Hogg, 2026).

Foothold and customers

The relevant early foothold includes people outside convenient rental-store locations who accepted postal delivery in return for access to a broad catalogue (Christensen Institute, n.d.). This project interprets those preferences as a trade-off between immediacy and access. It does not assume that every early customer was a low-income consumer, a nonconsumer or uninterested in film quality. The claim is specific to the customer needs and performance dimension being compared.

Performance and customer mapping

The poster uses immediacy of film access as its conventional performance indicator. Higher on the vertical axis means less waiting before viewing. Store rental offered same-day pickup when the film was available; early mail rental required delivery. The early customer label identifies people willing to accept a delay, while the mainstream label identifies viewers seeking same-day access. The lines are a qualitative schematic, not measured scores or a statistical market forecast. They illustrate the performance trade-off and later improvement without implying that catalogue breadth, price and viewing experience are identical.

Technological and commercial enablers

Disruptive innovation depends on enabling technology, a business model and a supporting value network (Christensen Institute, 2025). Applied here, the web catalogue, postal service and distribution facilities supported a subscription alternative to a retail-store network. The interpretation concerns how the service was delivered and sustained, alongside the technology used.

Part 1 History impact and critical analysis

Historical development

Netflix launched its online DVD-rental service in 1998 (Netflix, Inc., 2002a). Its subscription service began in September 1999 (Netflix, Inc., 2003). In June 2002, it announced ten regional distribution centres intended to reduce delivery times (Netflix, Inc., 2002b). A March 2002 company announcement describes CineMatch support for title selection (Netflix, Inc., 2002c). Netflix introduced streaming in 2007 (Netflix, Inc., 2019). Blockbuster's 2010 bankruptcy marks an important incumbent outcome (Christensen Institute, n.d.). In April 2023, Netflix announced that its last DVD shipments would be on September 29, 2023 (Sarandos, 2023).

Improvement and mainstream competition

Faster delivery reduced the initial waiting disadvantage. Streaming later changed the access method for supported titles and connected viewers. These developments help explain the move toward broader competition on convenience. The graph treats 2007 as a change in delivery capability, not proof that all customers switched immediately or that every Netflix feature was separately disruptive.

Social impact and trade-offs

Home ordering offered an alternative to travelling to a store. A larger accessible catalogue could serve viewers with fewer convenient local options (Christensen Institute, n.d.). The benefit depended on what the viewer valued. Postal delivery still required waiting, while streaming introduced dependence on connectivity and available content. Early streaming also had catalogue limitations; some customers valued the social experience of visiting stores (Arnett & Young, 2024).

Critical interpretation

The case supports a model of changing customer access and competition. It does not establish that Netflix alone caused Blockbuster's bankruptcy or that the conceptual graph measures market share. Using one performance indicator makes the trade-off clear but cannot represent every dimension of a rental service. Historical outcome and model interpretation should therefore remain distinct.

Connection to Part 2

Netflix's early service included title-selection assistance (Netflix, Inc., 2002c). The C++ programme illustrates this catalogue-discovery benefit through explicit preference rules. Its fictional catalogue and fixed weights make the decision process explainable. It does not recreate CineMatch, operate a streaming service or use Netflix data.

Part 1 References

- Arnett, T., & Young, K. (2024, July 26). *Taking K-12 education transformation from pipe dream to pipeline*. Christensen Institute.
<https://www.christenseninstitute.org/blog/taking-k-12-education-transformation-from-pipe-dream-to-pipeline/>
- Christensen Institute. (n.d.). *Disruptive innovation theory*.
<https://www.christenseninstitute.org/theory/disruptive-innovation/>
- Christensen Institute. (2025, September 24). *What is disruptive innovation? [Video]*.
<https://www.christenseninstitute.org/video/what-is-disruptive-innovation/>
- Hogg, A. S. (2026, March 18). *No, disruption isn't a strategy*. Christensen Institute.
<https://www.christenseninstitute.org/blog/no-disruption-isnt-a-strategy/>
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<https://ir.netflix.net/investor-news-and-events/financial-releases/press-release-details/2002/Netflix-Announces-Initial-Public-Offering/default.aspx>

Part 1 References continued

- Netflix, Inc. (2002b, June 20). *Netflix announces opening of 10 regional distribution centers*.
<https://ir.netflix.net/investor-news-and-events/financial-releases/press-release-details/2002/Netflix-Announces-Opening-of-10-Regional-Distribution-Centers/default.aspx>
- Netflix, Inc. (2002c, March 6). *Netflix files registration statement for initial public offering*.
<https://ir.netflix.net/investor-news-and-events/financial-releases/press-release-details/2002/Netflix-Files-Registration-Statement-for-Initial-Public-Offering/default.aspx>
- Netflix, Inc. (2003). *Annual report on Form 10-K for the fiscal year ended December 31, 2002*. U.S. Securities and Exchange Commission.
<https://www.sec.gov/Archives/edgar/data/1065280/000095016803001155/d10k.htm>
- Netflix, Inc. (2019). *Annual report on Form 10-K for the fiscal year ended December 31, 2018*.
https://s22.q4cdn.com/959853165/files/doc_financials/annual_reports/2018/Form-10K_Q418_Filed.pdf
- Sarandos, T. (2023, April 18). *Netflix DVD - The final season*. Netflix.
<https://about.netflix.com/en/news/netflix-dvd-the-final-season>

Part 2 Programme design

Purpose and integration

The Movie Discovery Assistant recommends a fictional film based on genre, mood and preferred duration. It illustrates one customer-facing function associated with a searchable online catalogue: finding a title suited to a user's preferences. It runs offline without login or subscriber information.

Inputs and outputs

Genre choices are Sci-Fi, Comedy and Drama. Mood choices are Relaxed, Exciting and Thoughtful. Duration choices are up to 100 minutes and over 100 minutes. The output contains the selected film's title, duration, description and the preferences it actually matches. A repeat menu lets the user search again or exit.

Selection rule

Genre is a required filter. Within that genre, a mood match adds two points and a duration match adds one point. The highest-scoring film wins; equal scores retain the first title in catalogue order. Mood therefore has priority over duration. A fallback stays within the selected genre and never claims an unmatched preference. A guard handles the absence of titles if the catalogue is later changed.

Input handling and implementation

C++17 is used with a `Movie` structure, a vector of records and a reusable `readChoice` function. `getline` reads the entire input. An `istringstream` parses a bounded integer and checks for trailing non-whitespace characters. Text, decimals, blank input, overflow and out-of-range entries are rejected. End-of-input exits cleanly. The programme uses conditional statements and loops to produce its recommendation.

Limitations

The catalogue contains 12 fictional titles. Weights are manually chosen and do not learn from preferences or ratings. No user-effectiveness evaluation, account features, persistence or external catalogue integration is included. The automated tests verify expected programme behaviour, rather than the real-world usefulness of recommendations.

Build: `g++ -std=c++17 -Wall -Wextra -pedantic src/main.cpp -o movie_assistant`

Part 2 Code capture input validation

This extract is generated from the actual source file. The complete source is included in src/main.cpp. Whole-line parsing rejects numeric prefixes followed by letters or decimals.

```
32 int readChoice(const std::string& prompt, int minimum, int maximum) {
33     while (true) {
34         std::cout << prompt;
35         std::string line;
36         if (!std::getline(std::cin, line)) {
37             std::cout << "\nInput ended. Goodbye.\n";
38             std::exit(0);
39         }
40         std::istringstream input(line);
41         int choice = 0;
42         char extra = '\0';
43         if (input >> choice && !(input >> extra)
44             && choice >= minimum && choice <= maximum) {
45             return choice;
46         }
47         std::cout << "Please enter a whole number from " << minimum
48                 << " to " << maximum << ".\n";
49     }
50 }
```

Figure 1. Actual readChoice implementation with source line numbers.

Part 2 Code capture recommendation logic

Genre is required. Mood contributes two points and duration contributes one. Equal scores keep the earlier catalogue entry. The null-result guard protects the output if the catalogue is later changed.

```
66     int bestScore = -1;
67     const Movie* best = nullptr;
68     for (const Movie& movie : catalogue) {
69         if (movie.genre != genres[genre - 1]) continue;
70         int score = 0;
71         if (movie.mood == moods[mood - 1]) score += 2;
72         if ((movie.minutes <= 100) == (length == 1)) score += 1;
73         // Equal scores keep the first title in catalogue order.
74         if (score > bestScore) {
75             best = &movie;
76             bestScore = score;
77         }
78     }
79     // Protect the output if a genre is removed from a future
catalogue.
80     if (best == nullptr) {
81         std::cout << "No titles are available in this genre.\n";
82         again = readChoice("Find another? 1 Yes 2 No: ", 1, 2) == 1;
83         continue;
84     }
```

Figure 2. Scoring, tie handling and missing-catalogue guard from the actual source.

Part 2 Actual programme runs

The following capture comes from a real interactive terminal run of the compiled programme. Typed menu values are visible. It demonstrates repeated use and the recommendations Coffee and Clouds and Orbit Station.

```
MOVIE DISCOVERY ASSISTANT
A small rule-based demonstration inspired by online movie discovery.
The titles below are fictional; no account or internet is needed.

Genre: 1 Sci-Fi  2 Comedy  3 Drama
Choose genre (1-3): 2
Mood: 1 Relaxed  2 Exciting  3 Thoughtful
Choose mood (1-3): 1
Length: 1 Up to 100 min  2 Over 100 min: 2

Recommendation: Coffee and Clouds (118 min)
Old friends meet again in a tiny hillside cafe.
Why: It matches your Comedy choice, Relaxed mood, and length preference.
Find another? 1 Yes  2 No: 1

Genre: 1 Sci-Fi  2 Comedy  3 Drama
Choose genre (1-3): 1
Mood: 1 Relaxed  2 Exciting  3 Thoughtful
Choose mood (1-3): 3
Length: 1 Up to 100 min  2 Over 100 min: 1

Recommendation: Orbit Station (96 min)
A crew solves a mystery aboard a research station.
Why: It matches your Sci-Fi choice, Thoughtful mood, and length preference.
Find another? 1 Yes  2 No: 2
Thanks for exploring.
```

Figure 3. Actual terminal output and entered values. The full transcript is assets/program_output.txt.

```
MOVIE DISCOVERY ASSISTANT
A small rule-based demonstration inspired by online movie discovery.
The titles below are fictional; no account or internet is needed.

Genre: 1 Sci-Fi  2 Comedy  3 Drama
Choose genre (1-3): 1abc
Please enter a whole number from 1 to 3.
Choose genre (1-3): 1.5
Please enter a whole number from 1 to 3.
Choose genre (1-3): 1
```

Figure 4. Actual rejection of 1abc and 1.5 before valid input. Full evidence is assets/invalid_output.txt.

Part 2 Testing and results

35 black-box checks passed: 18 preference combinations, 8 invalid inputs, 4 EOF cases, repeat, whitespace, menu bounds, fallback and mood priority.

The test driver executes the compiled programme as a separate process and checks explicit expected outputs. The complete 35-case record is `assets/test_results.csv`. These checks validate behaviour; they do not measure recommendation usefulness.

Case	Input	Expected	Actual	Status
C01	1, 1, 1; exit 2	Orbit Station	Orbit Station	PASS
C08	2, 1, 2; exit 2	Coffee and Clouds	Coffee and Clouds	PASS
C17	3, 3, 1; exit 2	River Letters	River Letters	PASS
I01	'wrong'; then 1,3,1,2	Reject and recover	Rejected and recovered	PASS
I05	'1abc'; then 1,3,1,2	Reject and recover	Rejected and recovered	PASS
I06	'1.5'; then 1,3,1,2	Reject and recover	Rejected and recovered	PASS
I07	"; then 1,3,1,2	Reject and recover	Rejected and recovered	PASS
E01	"	Clean EOF exit	Clean EOF exit	PASS
R01	2,1,2,1; 1,2,1,2	Two recommendations and normal exit	Two recommendations and normal exit	PASS
B01	invalid length 3 and repeat 7	Reject at both menus and recover	Rejected at both menus	PASS
F01	Sci-Fi, Relaxed, short	Fallback does not claim a mood match	No false mood match	PASS
P01	Drama, Exciting, short	Mood priority; no false length claim	Night Shift; no length claim	PASS

Test command: `sh test.sh`. The test runner requires Python 3; the C++ programme itself does not.

Part 2 Git development record

The log below is generated from the actual repository used to prepare this package. Original commits identify Codex draft; revision commits identify Codex assisted revision. They document AI-assisted preparation, not named students' contributions. The group must record genuine contributions through its own actual review and commits.

```
* ff2cf6b Document verified Netflix lifecycle model and six-member submission materials
* 27ea971 Validate complete menu lines and verify recommendation and recovery scenarios
* a4ee86f Clarify foothold stage and add presentation guidance and disclosure
* 41caa9c Lay out A3 poster and report with readable code capture
* f31ad38 Capture actual program run and local development history
* af73f2b Handle invalid and ended input; add five scenario checks
* 0270ef4 Implement menu inputs and fictional film catalogue
* eddae02 Document scope and learning goals for the movie assistant
```

Figure 5. Real git log --oneline --graph --all, captured before the final artifact-packaging commit. Full author names and timestamps for this snapshot are in assets/git_history_authors.txt. The bundle also retains any later packaging commit.

The development_history.bundle restores the repository, source and history. The README explains git clone and the build/test commands. The brief's --online option is a typographical error; --oneline is the valid command.

Code and Git history access link: [INSERT WORKING SOURCE CODE AND GIT HISTORY URL]

Presentation links and assistance disclosure

Recorded group presentation on OneDrive: [INSERT WORKING ONEDRIVE VIDEO URL]

Source code and Git history: [INSERT WORKING SOURCE CODE AND GIT HISTORY URL]

Turnitin reports: [ATTACH REQUIRED REPORTS OR INSERT ACCESS LINK]

Speaker	Suggested section	Approximate time
Aziel Tan Zheng Chuan	Case and project introduction	1:30
See Wing Kit	Model and customer mapping	2:00
Soo Kian Rong	History and impact	2:00
Vincent Lock Chun Kit	Programme design and logic	2:30
Wong Kee Yuan	Live demo and validation	3:00
Ho Ming Hao	Testing Git and conclusion	2:30

The complete six-person script is provided separately. The suggested total is about 13:30, allowing rehearsal and transitions within a target of 14-16 minutes. The actual recording must not exceed 18 minutes and must use MPEG, MP4 or MOV format.

Generative AI assistance disclosure

OpenAI ChatGPT (Codex) assisted with initial preparation on 25 September 2026 and revision on 30 September 2026, including analysis, source verification, poster/report layout, code, tests, evidence and script. The Git record identifies that assistance. No student contributions, signatures, completed video or Turnitin result are asserted by these records. Full disclosure is included in AI_DISCLOSURE.txt.

Each member must enter their actual contribution, personally complete their declaration and follow the course's originality and AI-content requirements before submission.